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Research on the Rule System, Game Mechanism, and Scoring Model of Chinese Local Mahjong Variants: A Case Study of the "Wanghuzhuang" Village

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Wanghuzhuang Village.

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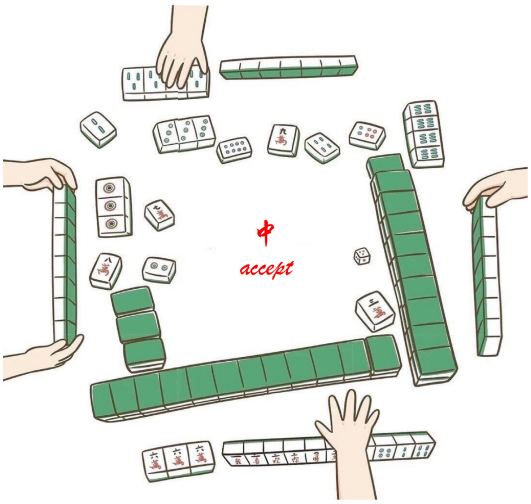

Abstract: This paper systematically documents the rule system and scoring model of "Wanghuzhuang" Mahjong, a unique Chinese local variant. Through field investigation and rule deconstruction, the study details seating determination, tile composition, core winning preconditions (the "Qingmen" requirement and "≥8 tiles restriction), special hand patterns (Seven Pairs, Dragon Hand), and the dynamic "Hui'er" wildcard mechanism. It analyzes the zero-sum scoring system: initial 15-point allocation, differentiated scoring for standard models (self-draw only) versus special patterns (allowing dianpao), "Qingyise" doubling, and the "zero-point protection" clause. The study finds that "Wanghuzhuang" fundamentally reshapes player strategies through its mandatory "Qingmen" precondition and "Hui'er" uncertainty, constructing a multi-layered risk-reward model. The "zero-point protection" clause embodies inclusive folk wisdom ensuring game integrity. This research concludes that "Wanghuzhuang" Mahjong constitutes a structurally rigorous, logically self-consistent system rich in strategic variables, offering a vivid sample for studying folk wisdom, probability calculation, and decision-making psychology.

Keywords: Wanghuzhuang Mahjong, Game Mechanism; Qingmen, Scoring Model, Zero-Sum Game

1. Introduction

As a strategic game prevalent across East Asia and globally, Mahjong exhibits a highly pronounced regional diversity in its rule systems. Localized variants often embody specific community cultures and gaming wisdom. The "Wanghuzhuang" Mahjong examined in this study represents a four-player variant circulating within a particular region, which has developed a distinctive set of victory conditions, dynamic gaming mechanisms, and a sophisticated scoring system built upon traditional Mahjong foundations. Currently, academic research on specialized studies of specific local Mahjong rules remains sparse, with quantitative analysis of their scoring models particularly underrepresented. This study aims to comprehensively deconstruct the rule architecture and scoring system of "Wanghuzhuang" Mahjong through standardized academic discourse for the first time, examining its unique "Qingmen" restrictions, the dynamic mechanism of "Hun'er" tiles, the gaming value of special hand patterns, and the zero-sum game logic underlying score distribution. The objective is to provide a detailed documentary reference for the study and preservation of folk gaming rules.

2. Methodology

This study employs rule-text analysis and participant observation. The researcher first collected and organized detailed oral and written rules concerning "Wanghuzhuang" Mahjong, subsequently systematizing fragmented rule entries into an internally consistent game-theoretic model through logical deduction. The exposition herein strictly follows the temporal sequence of gameplay: from preparatory procedures, through game progression, to terminal adjudication and score settlement, with systematic analysis at each stage.

3. Preparatory Procedures and Seating Determination

The opening of a "Wanghuzhuang" Mahjong session comprises two critical steps: orientation determination and dealer designation, which are integrated through two sequential stochastic events.

3.1 Orientation Determination Mechanism

At the commencement of play, the four players' directional positions, namely East, South, West, and North, must be established randomly. The procedure is as follows:

Determination of East: Any player simultaneously casts two dice. Taking their own position as the starting point, positions are counted counter-clockwise; the position indicated by the sum of the dice points is designated as "East". The correspondence is as follows: Firstly, sum of 5 or 9: The dice-caster themselves is East. Seconely, sum of 2, 6, or 10: The player immediately counter-clockwise to the dice-caster (their "lower house") is East. Thirdly, sum of 3, 7, or 11: The player opposite the dice-caster (their "facing house") is East. Lastly, sum of 4, 8, or 12: The player immediately clockwise to the dice-caster (their "upper house") is East.

Lottery Positioning: Upon determination of the "East" position, the remaining "South", "West", and "North" positions are naturally established in counter-clockwise sequence. Subsequently, the four players randomly draw from four pre-prepared directional tiles to finalize their respective seating positions.

3.2 Tile Composition

The game utilizes a standard Mahjong set

comprising 136 tiles, with the following composition:

Honor tiles: East, South, West, North, Red Dragon, Green Dragon, White Dragon (4 tiles each, totaling 28 tiles)

Suited tiles: Characters (Wan) 1 ~ 9, Circles (Tong) 1 ~ 9, Bamboos (Tiao) 1 ~ 9 (4 tiles each, totaling 108 tiles)

The tile wall is constructed by the four players collectively, with each player stacking 17 piles (pronounced dūn) in front of their position; each pile contains 2 tiles, totaling 34 tiles per player.

3.3 Initial Score Allocation

Each player receives an initial score of 15 points prior to gameplay commencement. Scores possess non-negativity; that is, a player's score may decrease to zero but cannot become negative. This stipulation constitutes the foundation for the subsequent "zero-point protection" clause within the scoring rules.

4. Gameplay Sequence and Core Mechanisms

4.1InitialDraw(Opening the Wall)

The game is initiated by the player occupying the East position (the inaugural dealer).

Dice Casting for Position: The dealer first casts two dice; based upon the sum of the points, positions are counted counter-clockwise to identify the corresponding player. This player then casts two dice a second time.

Opening the Wall: The sum of the two dice casts is aggregated to determine the wall-breaking position. Starting from the wall in front of the player "identified" by the dealer, the dealer counts from right to left (clockwise) to the pile number corresponding to this aggregate sum, skips that pile, and begins drawing from the subsequent pile.

Drawing Sequence: Players draw in the order of East, South, West, and North. Each player successively draws two piles (4 tiles), repeating this cycle three times, with each player obtaining 12 tiles in total.

Jump Draw: Upon completion of the three drawing cycles, the dealer (East) executes a "jump draw", which entails drawing the upper tile of the first remaining pile and the upper tile of the third remaining pile from the current wall, totaling 2 tiles.

Supplementary Draw: Subsequently, South draws the lower tile of the first pile, West draws the upper tile of the second pile, and North draws the lower tile of the second pile, each obtaining 1 tile.

"Hun'er" Determination: At this juncture, the lower tile of the first remaining pile (i.e., the third pile from the original sequence) is exposed and placed at the tail of the wall. This tile is designated the "indicator tile"; the subsequent tile in sequence (or its logical correspondence for honor tiles) determined by its face value becomes the "Hun'er" (wild tile) for the current hand. The rule stipulates: the indicator tile itself does not participate in gameplay; all four tiles of the "next" tile following it in sequence (cycling through numerals, e.g., following 5-Characters is 6-Characters; for honor tiles, the sequence typically cycles East-South-West-North-Red-Green-White) constitute the "Hun'er". In the 7-Hun'er variant, both the indicator tile and its subsequent tile are included as "Hun'er".

At this point, each player holds 13 tiles; the dealer, having drawn an extra tile through the jump draw, thus possesses 14 tiles and must immediately discard one superfluous tile, formally entering the tile-playing phase.

4.2 Playing Rules and Victory Prerequisites

The most distinctive characteristic of "Wanghuzhuang" Mahjong lies in its stringent

victory prerequisites, which constitute the core of its strategic gameplay.

Qingmen: In the process of pursuing a winning hand, a player must completely discard all tiles of one suit among the three suited categories, i.e., Characters (Wan), Circles (Tong), and Bamboos (Tiao). That is, at the moment of declaring a win, the player's hand may contain at most two suits of suited tiles (plus honor tiles).

Quantity Threshold: Upon satisfying the "Qingmen" condition, the player must hold a minimum of 8 tiles in any one remaining suit (or the honor tile collection) to obtain victory eligibility.

4.3 Winning Hand Models and Classification

Only upon satisfying the aforementioned prerequisites may a player declare a win according to the following models.

4.3.1 Standard Model

The most common winning hand form is the "3-3-3-3-2" model, wherein the hand comprises four "three-tile combinations" and one "pair". Under "Wanghuzhuang" rules, such standard hand patterns may only achieve victory through self-draw, locally termed "Ti" (pronounced dī). The three-tile combinations may be:

Sequence: Three tiles of the same suit with consecutive numerals.

Triplet: Three identical tiles.

Kong: Four identical tiles. Notably, this rule set does not permit "Pong" (claiming a discard to form a triplet), but does permit "Kong". Kongs are classified as exposed kong (konging a tile discarded by another player) and concealed kong (self-drawing all four identical tiles).

4.3.2 Special Hand Patterns

To enhance game variability and interest, "Wanghuzhuang" introduces two high-value special hand patterns that permit "dianpao" (point-and-cannon), that is, winning off a tile discarded by another player.

Seven Pairs: The hand comprises seven pairs.

Dragon Hand: The hand contains a "one-through-nine" sequence (a complete series of 1~9) of a single suit (constituting 9 tiles), supplemented by one "three-tile combination" and one "pair", forming a "9+3+2" model. This constitutes a high-difficulty hand pattern.

4.3.3 Qingyise

When a player's winning hand is composed entirely of a single suit (all Characters, or all Circles, or all Bamboos, excluding honor tiles), it constitutes "Qingyise" (Pure Straight). As a hand pattern modifier, Qingyise doubles the final settlement score.

5. Kong Rules and Scoring System

The scoring system of "Wanghuzhuang" follows strict temporal logic: kong settlement precedes win settlement. Score transfers adhere to zero-sum game principles—the sum of points gained by winners equals the sum of points lost by losers.

5.1 Kong Rules and Dynamic Adjustment of the Wall Tail

Kong Proceeds:

Exposed Kong: The kong-declaring player collects 1 point from each of the other three players, totaling 3 points in income.

Concealed Kong: The kong-declaring player collects 2 points from each of the other three players, totaling 6 points in income.

Post-Kong Draw: Upon declaring a kong, the player must immediately supplement their hand by drawing one tile from the tail of the wall.

Dynamic Adjustment of the Wall Tail: Each kong declaration shifts the starting position of

the "dead wall" (non-playing tile area) one pile toward the tail. *Specifically:*

No kongs: The final 7 piles do not participate in play.

1st~2nd kongs: The 2nd through 8th piles from the end do not participate in play.

3rd kong: The 3rd through 9th piles from the end do not participate in play, and so forth.

This rule ensures that each kong not only generates immediate score proceeds but also accelerates the termination of the hand.

5.2 Win Scoring Rules

5.2.1 Standard Model Scoring (Self-Draw Only)

When a player wins by self-draw with the standard "3-3-3-3-2" model, score transfers proceed as follows:

Losers (the other three players) each forfeit 2 points.

The winner (self-drawer) gains 2 points from each loser, totaling 6 points in income.

5.2.2 Special Hand Pattern Scoring

Special hand pattern scoring distinguishes between self-draw and dianpao (winning off a discard):

Self-Draw: Losers each forfeit 6 points; the winner gains 6 points from each loser, totaling 18 points in income.

Dianpao: Losers each forfeit 3 points; the winner gains 3 points from each loser, totaling 9 points in income.

5.3 Qingyise Doubling Rule

If the winning hand satisfies the "Qingyise" condition, all aforementioned win scores are multiplied by 2 based on the base value. Specifically: *Standard Model Qingyise Self-Draw:* Losers each forfeit 4 points (2×2); winner gains 12 points. Special Pattern *Qingyise Self-Draw:* Losers each forfeit 12 points (6×2); winner gains 36 points. Special Pattern Qingyise Dianpao: Losers each forfeit 6 points (3×2); winner gains 18 points.

5.4 Zero-Point Protection Clause

Given the non-negativity of scores, when a player's score decreases to 0, the "zero-point protection" mechanism is triggered:

Players with 0 points are exempt from paying any points to winners or kong-declarers.

Such players retain the right to collect their entitled points from other players (that is, if the player is a winner or kong-declarer, they may receive points normally).

This clause ensures game continuity, preventing player elimination due to score depletion, while simultaneously preserving the possibility of comeback for score-trailing players.

6. Dealer Rotation Mechanism

The game adopts the standard four-round system of East, South, West, and North. The inaugural hand is dealt by East wind. If the dealer wins the hand, they continue to "sit as dealer" (retain the dealership). If the dealer is defeated by any other player or the hand ends in an exhaustive draw, the dealer "descends from the dealership", and the next player in counter-clockwise sequence ascends as the new dealer. When the game progresses to the fourth round (North round) and the North wind seat is defeated or results in an exhaustive draw, the game concludes in its entirety. Dealer rotation itself does not generate any score fluctuations.

7. Discussion and Conclusions

As a localized variant, "Wanghuzhuang" Mahjong demonstrates innovative evolution within standardized frameworks through its rule design and scoring model.

7.1 Multi-Layered Strategic Gameplay

This study reveals that its core gaming logic does not merely pursue rapid winning

declaration; rather, it necessitates structural hand planning under the dual rigid constraints of "Qingmen" and "≥8 tiles". This compels players to establish their primary suit development direction early in the game and decisively discard other suits, substantially increasing strategic depth. The introduction of the scoring system further reinforces this characteristic: pursuing high-yield special hand patterns (Seven Pairs/Dragon Hand) entails assuming longer formation cycles and higher risks, while the stable returns of the standard model provide conservative players with a safety margin.

7.2 Zero-Sum Game and Risk Equilibrium

The scoring system exhibits typical zero-sum game characteristics. The immediacy of kong proceeds and the terminal nature of win proceeds generate temporal-dimensional gaming tension. The settlement sequence of "kong before win" signifies that kongs not only yield immediate proceeds but may also influence final win attribution through altering wall structure (draw mechanics and dead wall adjustment). This design compels players to weigh short-term gains against long-term winning probability.

7.3 Sociological Significance of the Zero-Point Protection Clause

The design of the "zero-point protection" clause embodies "inclusive" wisdom within folk gaming rules. It prevents the "Matthew Effect" within the scoring system, ensuring that even players in absolute disadvantageous positions can continue participating in the game and retain theoretical comeback opportunities, thereby maintaining the integrity and social enjoyment of the four-player game until the final conclusion.

Notes

The author declares no competing financial interest.

References

No references were cited in this study.